

Amortized PEFT-Conditioned Data Selection for Cross-PEFT Fine-Tuning

Anonymous submission

Abstract

Parameter-efficient fine-tuning (PEFT) is rarely deployed as a single configuration: a lab or platform maintains a registry of LoRA ranks, module placements, adapters, (IA)³, and other recipes chosen for different tasks and deployment budgets. Gradient- and influence-based data selectors produce strong subsets but bind to the target PEFT—each new configuration rebuilds a target-specific gradient datastore—so selection cost grows linearly with the registry. Reusing one configuration’s selected subset removes that cost but sacrifices quality, because a subset optimized for one trainable subspace is mismatched to the others.

We present PCU-Select, which recasts this per-target recomputation as reusable supervision. It models example utility as a quantity conditioned jointly on the task and a structured PEFT code, builds a shared representation of samples, tasks, and PEFT configurations once offline, and trains a conditional scorer with multi-fidelity utility labels. Transferring to a new target PEFT requires only encoding its PEFT code and task sketch and one forward pass over cached sample features—no per-target gradient datastore.

Across Llama-2-7B, four tasks, and five seen PEFT configurations, PCU-Select improves consistently over PEFT-agnostic selectors and near-ties a per-PEFT LESS reimplementation on aggregate downstream quality while still producing distinct per-target subsets. Its contribution is amortization, not a single-configuration win: after a one-time 72.0 GPU-hour offline supervision, each target costs 0.18 GPU-hours, which reduces five-PEFT selection compute from 158.0 (per-PEFT LESS) to 72.9 GPU-hours and breaks even after 2.3 targets. A single reused LESS subset, by contrast, costs a full LESS run yet falls below the PEFT-agnostic frontier. PCU-Select is therefore best read as a reusable selector for a PEFT registry; targets far from the seen registry still require light calibration.

1 Introduction

Adapting large language models under limited compute requires efficiency on both the model side and the data side. A lab or platform rarely maintains a single adaptation setting: it keeps a *registry* of parameter-efficient fine-tuning (PEFT) configurations—LoRA at several ranks and placements, adapters, (IA)³, prefix tuning, or BitFit—chosen for

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different tasks, latency targets, memory limits, and deployment constraints. PEFT freezes the backbone and trains a small set of task-specific parameters, reducing optimizer memory and deployment storage. Data-efficient fine-tuning is the complementary tool: it selects a compact subset from a larger instruction pool, reducing the number of examples and training steps. These two tools are usually applied independently: PEFT controls which parameters move, while data selection controls which examples drive the update.

Current data selectors rarely model this interaction. Quality filters, diversity heuristics, proxy-model scores, and gradient-similarity methods often assign each example a single value for a target task and then reuse the selected subset across adaptation methods. That assumption is convenient, but PEFT methods change the trainable subspace itself. LoRA injects low-rank additive updates into selected projections (Hu et al. 2022); adapters insert residual bottlenecks (Houlsby et al. 2019); and (IA)³ learns multiplicative activation scalings (Liu et al. 2022). An example whose gradient is aligned with a task inside one subspace need not be aligned inside another.

When the same pool must serve a whole registry, this omission becomes a cost problem. The strongest selectors are gradient- and influence-based: LESS (Xia et al. 2024) builds a target-specific gradient datastore and ranks candidates by their influence on a validation sketch. Because that datastore is tied to the target adapter’s own trainable parameters, every new configuration pays to rebuild it, and selection cost grows linearly with the number of PEFTs served. The obvious shortcut—run the selector once and reuse its subset for every configuration—removes the cost but forfeits the quality that motivated a gradient selector: a subset chosen for one trainable subspace is mismatched to the others, and we measure that reusing a single LESS subset falls below even a PEFT-agnostic representation baseline (Table 3). This is the reuse-versus-recompute dilemma at the center of the paper: existing methods are either expensive per target or cheap but PEFT-agnostic in effect. The expensive supervision should instead be computed *once* and reused across configurations.

Our motivation study shows that PEFT-specific utility changes with the trainable subspace. We estimate short-horizon utility for the same candidate pool under LoRA, adapter, and (IA)³ variants, using two task sketches and independent anchor/seed replicates. Figure 1 shows that within-

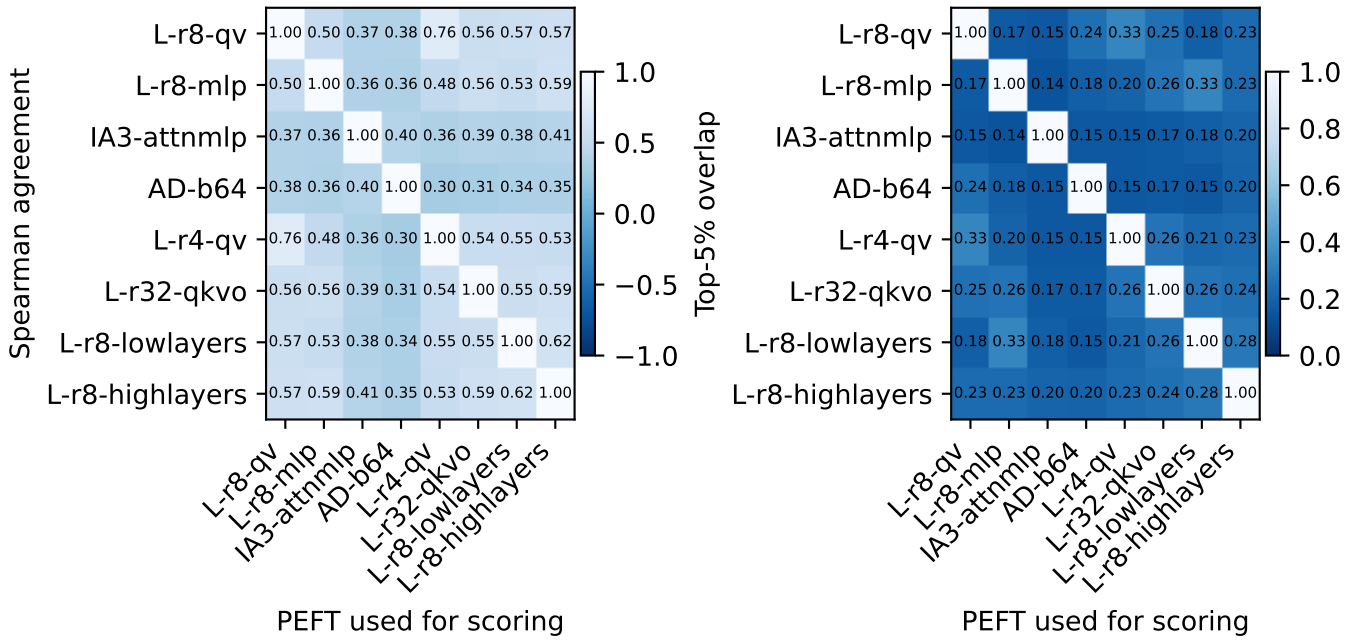


Figure 1: Short-horizon data-value estimates disagree across PEFT configurations. A common pool of 20K candidate examples is scored under each target PEFT with two task sketches and independent anchor/seed replicates; each cell is the Spearman rank correlation between the row and column scoring runs. The heatmap diagonal is self-correlation (1.00 by construction); *independent* same-PEFT replicates agree only moderately ($\rho = 0.79$, Top-5% overlap = 0.31), reflecting the short-horizon label-noise floor against which cross-PEFT agreement should be read. Cross-PEFT agreement falls further with structural distance: cross-family pairs average $\rho = 0.36$ and Top-5% overlap = 0.17.

PEFT agreement exceeds cross-PEFT agreement. Same-family capacity changes retain moderate structure, but placement changes and cross-family changes sharply reduce rank agreement and Top-5% overlap. A transfer matrix gives the downstream consequence: selecting data for the same PEFT used in training beats mismatched *cross-family* selection by 2.0 points on average across target PEFTs (Appendix Table 15); the Analysis section reports the narrower 1.45-point within-LoRA configuration-scan gap.

This paper treats data value as a PEFT-conditioned quantity. We ask: given a candidate example x , a target task t , and a target PEFT configuration p , can we predict whether x will help when the trainable parameters are exactly those exposed by p ? This framing differs from PEFT-agnostic selection, which asks whether x is generally good for the task. It also differs from recomputing an influence selector for every PEFT, because the expensive supervision should be reused across many target configurations.

PCU-Select (PEFT-Conditional Utility) addresses this question with an intervention-site utility layer. It maps samples, task sketches, and PEFT configurations into a shared set of 24 module-output sites in the backbone. Sample and task signatures describe gradient alignment at each site. The PEFT code describes which sites a method touches, how it acts there, and how much trainable capacity it has. PCU-Select uses a low-fidelity site-weighted gradient proxy for broad coverage, corrects it with short-horizon high-fidelity labels, and trains a conditional scorer that predicts both util-

ity and uncertainty. At deployment, the scorer ranks the pool for a target (p, t) and a quota allocator balances utility with semantic coverage.

We make three contributions.

- **Problem.** We formulate cross-PEFT data-efficient fine-tuning as an *amortization* problem: the goal is not to beat a per-PEFT influence selector on a single target but to preserve its quality while converting per-target recomputation into reusable supervision. We show empirically that data-value rankings differ across PEFT configurations (cross-family Top-5% overlap near 0.17, a 2.0-point downstream cross-family mismatch penalty), and that reusing one influence-selected subset therefore drops below even a PEFT-agnostic baseline.
- **Method.** We introduce PCU-Select, a reusable PEFT-conditioned selector built from intervention-site gradient signatures, structured PEFT encodings, task sketches, multi-fidelity utility labels, uncertainty-aware scoring, and adaptive cluster quotas.
- **Evaluation.** Across GSM8K, HumanEval, MMLU, and TyDiQA on a fixed Llama-2-7B backbone family, PCU-Select improves 10%-budget downstream quality over PEFT-agnostic selectors (over random by 2.89 points and RDS+ by 0.74 points) and near-ties a per-PEFT LESS reimplementation on the seen registry while still generating distinct per-target subsets. Its offline supervision amortizes after about 2.3 target configurations, reducing

five-PEFT selection compute from 158.0 to 72.9 GPU-hours ($2.2\times$).

2 Related Work

Instruction-data selection. Instruction tuning benefits from small but carefully selected data. LIMA shows that 1,000 curated examples can produce strong instruction-following behavior in a large model (Zhou et al. 2023). AlpacaGus filters low-quality Alpaca examples with LLM judgments (Chen et al. 2024); InsTag measures semantic diversity and complexity through instruction tags (Lu et al. 2023); and DEITA combines quality, complexity, and diversity criteria for automatic data selection (Liu et al. 2024). Other selectors emphasize explicit diversity through quality-diversity trade-offs, log-determinant objectives, or partition-aware sampling (Bukharin et al. 2023; Wang et al. 2024; Rao et al. 2024). These methods improve data efficiency, but they mostly treat value as a property of the example and target task rather than of the PEFT subspace used for adaptation.

Proxy-model and influence selection. Model-assisted selectors score examples with external judges, weak models, or training signals. Cherry LLM uses instruction-following difficulty (Li et al. 2023); Superfiltering studies when weak models can rank data for stronger models (Li et al. 2024); and SmallToLarge transfers trajectory information from small models to larger targets (Yang et al. 2024). Influence-style methods estimate how training examples affect validation behavior. Classical influence functions use second-order approximations (Koh and Liang 2017), TracIn accumulates gradient similarities along checkpoints (Pruthi et al. 2020), and LESS builds a low-dimensional gradient datastore for instruction selection (Xia et al. 2024). DataInf specializes influence approximation for LoRA-tuned generative models (Kwon et al. 2024). These methods recompute target-specific gradients or influence for every new configuration. PCU-Select instead conditions a learned scorer on a structured PEFT code, so one offline supervision serves many targets and the expensive per-target recomputation is amortized rather than repeated; we compare against both a per-PEFT LESS reimplement and a cheap shared-datastore Influence baseline.

Parameter-efficient fine-tuning. PEFT methods reduce adaptation cost by updating a small module or parameter subset. Adapters insert bottleneck modules into Transformer layers (Houlsby et al. 2019; Pfeiffer et al. 2021). Prefix and prompt tuning optimize continuous prompts or prefix states (Li and Liang 2021; Lester, Al-Rfou, and Constant 2021). LoRA learns low-rank additive updates (Hu et al. 2022), QLoRA combines quantization with LoRA for memory efficiency (Dettmers et al. 2023), BitFit tunes only bias terms (Ben Zaken, Goldberg, and Ravfogel 2022), and (IA)³ rescales attention and feed-forward activations (Liu et al. 2022). PEFT work usually compares accuracy, parameter count, memory, or inference overhead under fixed data. This paper asks a complementary question: when the candidate pool is fixed, do different PEFT methods assign high utility to the same examples?

Joint data- and parameter-efficient adaptation. Data selection and PEFT address two sides of efficient adaptation, but most work optimizes one side while holding the other fixed. DataInf and LESS move toward the intersection by using gradients from LoRA-style training (Kwon et al. 2024; Xia et al. 2024), and concurrent geometry-aware targeted selection suggests that PEFT optimization has nontrivial subspace structure (Min et al. 2026). These still tie the selector to one optimization geometry. PCU-Select treats the *structured PEFT configuration itself*—its sites, operator, capacity, and recipe—as an explicit condition of the selector. Unlike PEFT-agnostic filters, PCU-Select changes the selected subset when the trainable subspace changes; unlike per-PEFT influence methods, it amortizes the expensive supervision across target configurations.

3 Problem Formulation

We formalize *cross-PEFT data-efficient fine-tuning*: selecting a compact training subset whose value depends on the target PEFT configuration and target task. Fix a backbone family and frozen base model M_0 . A candidate pool $\mathcal{D} = \{x_i\}_{i=1}^N$ contains instruction-response examples. A task t provides a small validation sketch V_t drawn from train or development data, never from the test set. A PEFT configuration $p \in \mathcal{P}$ specifies the family, trainable locations, capacity, and training recipe. Given budget B , the ideal subset solves

$$S^* = \arg \max_{S \subseteq \mathcal{D}, |S| \leq B} \text{Perf}(\text{Tune}(M_0, p, S), V_t^{\text{test}}). \quad (1)$$

The held-out set V_t^{test} appears only in this definition of the oracle; it is never used for training or for selection, which see only the sketch V_t . Directly optimizing Eq. (1) is infeasible because it requires training and evaluating many subsets for every target (p, t) pair.

PCU-Select replaces the subset oracle with a conditional per-example utility,

$$s_\phi(x, p, t) \approx u(x, p, t), \quad (2)$$

where u measures how much x helps task t when the trainable parameters are those of p . Concretely, $u(x, p, t) \in [0, 1]$ summarizes sketch-loss reductions into a rank-normalized short-horizon utility score, as defined in the Method section. Conditioning on p is the core departure from PEFT-agnostic selection. A sample’s gradient can be useful in the attention projections trained by LoRA, weak in the residual bottlenecks trained by adapters, and different again under multiplicative (IA)³ gates. The selected subset should therefore change when the trainable subspace changes.

Amortization objective. A registry makes cost, not single-target accuracy, the operative criterion. Let $\mathcal{P}_{\text{target}} = \{p_1, \dots, p_T\}$ be the target PEFT configurations a fixed pool must serve. A per-target influence selector such as LESS pays a selection cost for each configuration, whereas an amortized selector pays a one-time offline cost plus a small online cost per target:

$$C_{\text{LESS}}(T) = T C_{\text{LESS}}^{\text{sel}}, \quad C_{\text{PCU}}(T) = C_{\text{offline}} + T C_{\text{PCU}}^{\text{sel}}, \quad (3)$$

with $C_{\text{PCU}}^{\text{sel}} \ll C_{\text{LESS}}^{\text{sel}}$. The goal is *quality-preserving amortization*: match the per-PEFT selector’s downstream quality,

$$\Delta_{\text{quality}} = \text{Perf}_{\text{PCU}} - \text{Perf}_{\text{LESS}} \approx 0, \quad (4)$$

while spending strictly less once the registry exceeds the break-even size $T^* = C_{\text{offline}} / (C_{\text{LESS}}^{\text{sel}} - C_{\text{PCU}}^{\text{sel}})$, i.e. $C_{\text{PCU}}(T) < C_{\text{LESS}}(T)$ for $T > T^*$. Under this criterion a near-tie with LESS is the design target, not a shortfall: LESS is the quality reference and PCU-Select is its amortized alternative. Reusing a single LESS subset is the degenerate point $C_{\text{offline}} = C_{\text{LESS}}^{\text{sel}}$, $C^{\text{sel}} = 0$ of the same trade-off—cheap, but PEFT-agnostic in effect and therefore quality-losing.

Task relativity is concrete: a worked-solution arithmetic example receives high utility under the GSM8K sketch but low utility under the TyDiQA sketch, because its gradient aligns with the reasoning task but not with multilingual span extraction; the same example’s utility also shifts across PEFTs when the aligned sites are not the ones a given method trains.

The formulation rests on four requirements. **PEFT dependence**: the scorer must take p as an explicit input. **Task relativity**: utility must be defined relative to V_t so that the model does not learn a generic quality score. **Structural encodability**: PEFT methods must be represented by their sites, operator type, capacity, and recipe, not just by a family name. **Amortizable cost**: expensive labels and scorer training are paid offline, while applying to a new (p, t) pair uses cached features and forward scoring.

4 Method: PCU-Select

PCU-Select learns the surrogate in Eq. (2) and uses it to select a budgeted subset for a target PEFT and task. Figure 2 summarizes the pipeline. The offline stage extracts reusable sample and task signatures, builds low- and high-fidelity utility labels, and trains the scorer. The online stage encodes a target (p^*, t^*) , checks whether p^* is outside the training configuration distribution, scores the pool, and allocates the budget across semantic clusters. Algorithms 1 and 2 in the appendix state both stages.

Intervention Sites

PCU-Select compares PEFT methods through module-output sites rather than raw parameter tensors. For a decoder-only Transformer, it selects eight evenly spaced layers and hooks three outputs in each layer: attention output, MLP output, and post-residual block output. The resulting set Ω has 24 sites. We use module outputs rather than parameter tensors because a module-output gradient is a linear image of the projection-level parameter gradient and is stable across backbones with fused or split projection implementations, so it approximates the update a PEFT would make there without tying the representation to a specific parameterization. Eight evenly spaced layers keep the cached signature small while covering the depth of the network, and retaining the post-residual block site is what lets adapters, which act on the residual stream, be represented in the same coordinate system as LoRA and (IA)³. We treat this site space as a motivated design choice and leave a full site-space ablation (layer

count, attention- vs. MLP-only sites, projection dimension) to future work.

Each PEFT configuration maps to a site mask and an operator type. Attention LoRA and (IA)³-attention touch attention-output sites; MLP LoRA and (IA)³-FFN touch MLP-output sites; adapters touch block-residual sites; prefix tuning touches attention-output sites; and BitFit is represented as a bias shift. PCU-Select assigns each site a normalized PEFT weight

$$\begin{aligned} w_p^\omega &= \text{mask}(p, \omega) \rho_{\text{op}} c(p, \omega), \\ c(p, \omega) &= \tanh(\eta \log(1 + \text{cap}(p, \omega) / d_{\text{model}})), \\ \tilde{w}_p^\omega &= w_p^\omega / \max(\epsilon, \sum_{\omega'} w_p^{\omega'}). \end{aligned} \quad (5)$$

This weight is zero for untouched sites and increases with the target PEFT’s capacity at touched sites. The capacity $\text{cap}(p, \omega)$ maps each family’s size knob—LoRA rank, adapter bottleneck width, (IA)³ vector count, prefix length, BitFit bias count—onto a common trainable-parameter scale (Appendix Table 12); the tanh saturates it so high-capacity configurations do not swamp the site weights, and the operator-weight prior ρ_{op} encodes that additive low-rank updates ($\rho=1.0$) exercise a site more directly than multiplicative gates (0.6), bottlenecks (0.8), prefixes (0.4), or bias shifts (0.3).

Representations and Utility Labels

A sample representation $z_x = [e_x; d_x; a_x] \in \mathbb{R}^{848}$ concatenates a 768-dimensional semantic embedding, 16 difficulty and format statistics, and a 64-dimensional activation signature. The semantic vector is the frozen `sentence-transformers/all-mpnet-base-v2` joint instruction-response embedding; we do not train this encoder on the candidate pool or target benchmarks. The difficulty vector contains instruction length, response length, total length, response/instruction ratio, seven response-LM statistics (mean/std/max loss, perplexity, average log-probability, mean/max entropy), and five format flags (reasoning marker, code marker, question marker, English, Chinese). The activation signature mean-pools response-token hidden states at the eight sampled layers and projects them to 64 dimensions.

A task sketch V_t is encoded to $z_t \in \mathbb{R}^{848}$ by pooling the same sample representation over 32 sketch examples and by averaging their site gradient signatures. A PEFT code $z_p = [m_p; c_p; r_p; f_p] \in \mathbb{R}^{192}$ concatenates a 24×4 site/operator mask (m_p , 96 dims), a 16-dimensional capacity vector (c_p), a 16-dimensional training-recipe vector (r_p), and a 64-dimensional family/operator embedding (f_p). The four mask channels are active-site, additive-update, multiplicative-scale, and prefix-style control bits. LoRA and adapters share the additive bit but differ through capacity and f_p ; BitFit uses the active bit plus the bias-shift prior, capacity vector, and family/operator embedding. The target-module set enters r_p through normalized module-count and placement features; at the 24-site granularity, q/v and q/k/v/o LoRA both touch attention-output sites, so these recipe and capacity features carry the projection-level distinction.

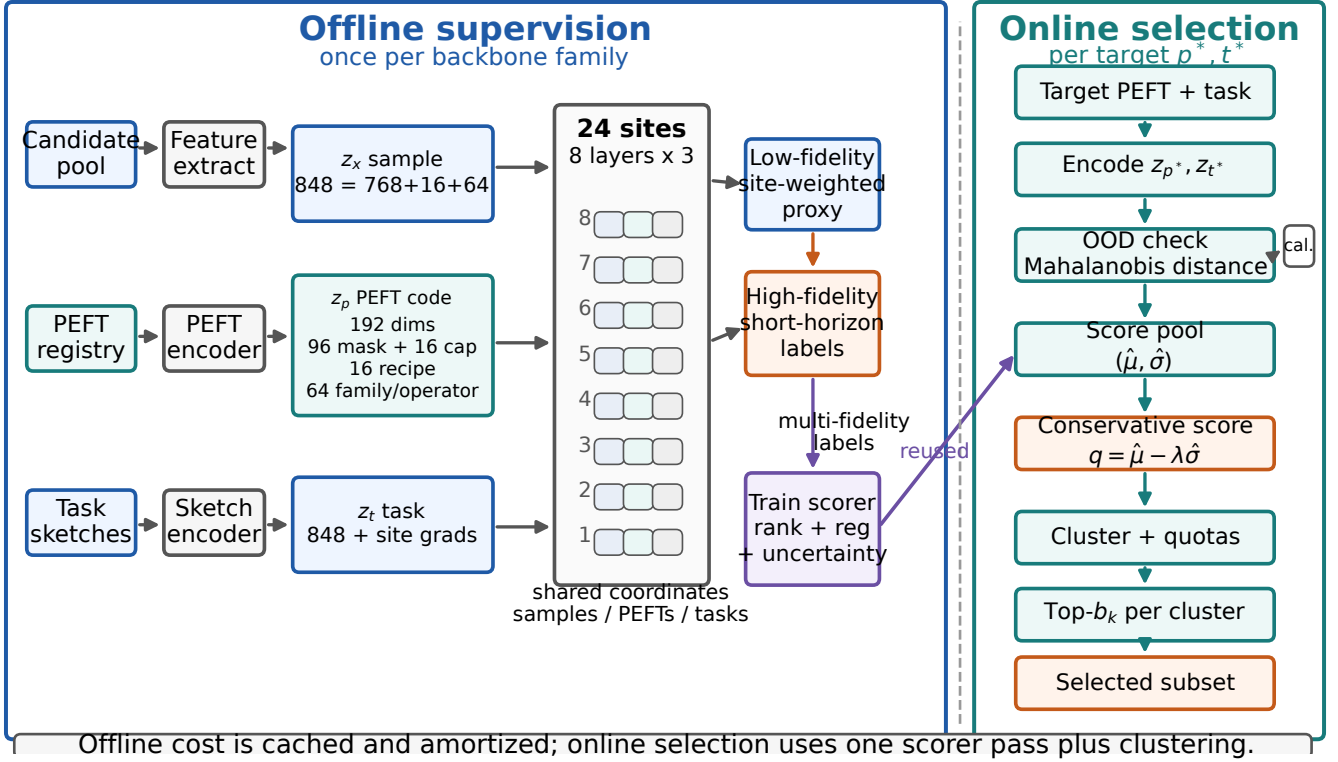


Figure 2: PCU-Select separates reusable offline supervision from cheap online selection. Intervention sites Ω provide the shared coordinate system that links sample gradients, task sketches, and PEFT configurations. The PEFT code is 192-dimensional: a 96-dimensional site/operator mask, 16 capacity dimensions, 16 recipe dimensions, and a 64-dimensional family/operator embedding.

For seen PEFTs, f_p is learned with the scorer. For Prefix, P-Tuning, and BitFit OOD targets, zero-shot scoring initializes f_p from the closest operator centroid (prefix-style or bias-shift) rather than from a random vector; calibration then fits the residual head described below. This makes zero-shot OOD a structured extrapolation, while the calibrated results measure how quickly a small target-specific label set repairs that extrapolation.

Low-fidelity utility aligns each sample with the target sketch on the sites trained by the target PEFT:

$$u^{\text{lo}}(x, p, t) = \sum_{\omega \in \Omega} \tilde{w}_p^\omega \cos(g_x^\omega, g_t^\omega). \quad (6)$$

Here g_x^ω is a unit-normalized random projection of the response-token loss gradient at site ω , and g_t^ω is the sketch mean. All task sketches use the same response-token cross-entropy for signatures; task-native metrics such as option likelihood, Pass@1, and F1 are used only for downstream evaluation. The cosine proxy can be negative, so PCU-Select rank-normalizes u^{lo} within each (p, t) bucket before proxy distillation and ranking losses. The proxy is cheap after sample signatures are cached, so it labels many (x, p, t) triples.

High-fidelity utility corrects the proxy near the selection boundary. For a sampled triple, PCU-Select performs short PEFT updates from a base anchor and a warm anchor, then

measures sketch-loss reduction:

$$\Delta(x, p, t, a, h) = L_{V_t}(\theta_a) - L_{V_t}(\text{Adapt}_p^h(x; \theta_a)). \quad (7)$$

Rank-normalized reductions from horizons $h \in \{1, 4\}$ and both anchors define u^{hi} . The high-fidelity budget is 10K triples, split across coverage sampling, uncertainty sampling, and boundary sampling. Each high-fidelity update uses the sampled example plus seven in-bucket fillers; labels therefore estimate a stable mini-batch marginal, not an isolated single-example gradient. Coverage and uncertainty labels are generated first, a provisional scorer is trained, and boundary sampling then targets scorer-vs-label rank disagreements from that provisional pass. Rank normalization is always within the same PEFT, task, anchor, and horizon bucket.

Scorer and Selection

The scorer predicts $(\hat{\mu}, \hat{\sigma}) = s_\phi(z_x, z_p, z_t)$. Three encoders map z_x , z_p , and z_t into hidden vectors. The PEFT and task codes modulate the sample code through FiLM, while a bilinear term models sample-PEFT interaction. Training combines high-fidelity ranking and regression losses, low-fidelity proxy distillation, and a heteroscedastic uncertainty loss. Stage A pretrains on the proxy for broad conditional structure; stage B enables all losses and selects checkpoints on held-out PEFT/task pairs.

At application time, PCU-Select ranks examples by a conservative score

$$q(x) = \hat{\mu}(x, p^*, t^*) - \lambda \hat{\sigma}(x, p^*, t^*), \quad \lambda = 0.2. \quad (8)$$

Pure top- k ranking can over-select near-duplicates, so PCU-Select clusters examples in semantic space and allocates budget across clusters:

$$b_k = \text{round} \left(B \frac{(v_k^+)^{\gamma} |C_k|^{1-\gamma}}{\sum_j (v_j^+)^{\gamma} |C_j|^{1-\gamma}} \right), \quad \gamma = 0.6. \quad (9)$$

Here v_k is the mean conservative score of the top 10% examples in cluster C_k , and $v_k^+ = \max(v_k, 0)$. For the 300K pool, $k = \max(50, \lfloor \sqrt{N} \rfloor)$ yields 547 clusters with about 548 examples per cluster on average; clustering is included in the 0.18 selection GPU-hours reported below. The final subset takes the top- b_k examples by q inside each cluster. Clusters with non-positive value ($v_k^+ \leq 0$) receive no budget, rounding remainders are distributed greedily by descending cluster weight, and any over-allocation beyond a cluster’s size is trimmed and reassigned, so the quotas always sum to exactly B . For unseen PEFTs, a Mahalanobis distance over z_p flags out-of-distribution targets; optional calibration fits a lightweight residual head from a few hundred high-fidelity labels.

5 Experiments

The evaluation asks four questions: does PCU-Select improve selected-data performance, does it amortize cost across PEFTs, which components matter, and how does it behave on unseen configurations?

Setup

We evaluate on four main tasks: GSM8K, HumanEval, MMLU, and TyDiQA. Metrics use each task’s native scale: exact match, Pass@1, accuracy, or F1. The candidate pool contains 300K mixed instruction examples, and the main selection budget is 10%; budget sensitivity also uses 5% and 30%. Each method trains the same target backbone, PEFT recipe, number of steps, and evaluation protocol; only the selected subset changes. Reported downstream values average over three target-training seeds.

The seen PEFT set contains attention LoRA, MLP LoRA, (IA)³, and adapters; unseen tests vary rank, placement, learning rate, and family. Baselines include Random and Balanced-Random; heuristic selectors based on length, loss, and perplexity; representation selectors such as embedding-nearest-neighbor, diversity clustering, and RDS+; and two gradient/influence selectors, Influence and per-PEFT LESS. Balanced-Random (cluster-stratified uniform sampling) scores 32.90 on average, within seed noise of plain Random, so the main table focuses on the strongest representative baselines and the appendix reports the omitted aggregate rows. Three baselines deserve precise definition because their names invoke prior work; the appendix gives full specifications. **RDS+** ranks candidates by cosine similarity between per-feature-standardized joint semantic embeddings

Table 1: Representative PEFT configurations. The five *seen* configurations are the scorer’s training support and the columns of Table 2; two representative *unseen* configurations are shown here, and the complete 17-configuration registry (seen, unseen-config, and OOD-family) used in the calibration study in the Analysis section appears in Table 11. Trainable-parameter counts and the touched fraction of the 24 intervention sites come from the registry counter for the Llama-2-7B backbone (6.74B parameters).

Config	Group	Family	Params	% bb	Sites
L-r8-qv	Seen	LoRA	4.19M	0.062	8/24
L-r16-qkvo	Seen	LoRA	16.78M	0.249	8/24
L-r8-mlp	Seen	LoRA	7.73M	0.115	8/24
IA3-attnmlp	Seen	IA ³	0.61M	0.009	16/24
AD-b64	Seen	Adapter	16.78M	0.249	8/24
L-r4-qv	Unseen	LoRA	2.10M	0.031	8/24
L-r32-qkvo	Unseen	LoRA	33.55M	0.498	8/24

(from a *frozen external* sentence encoder shared by all methods and independent of PCU-Select training) and the task-sketch mean; it is task-conditioned but PEFT-agnostic. **Influence** scores each candidate by cosine between its projected base-model gradient and the pooled sketch gradient over a *single shared* datastore—the cheap, PEFT-agnostic gradient baseline. **LESS** is our per-PEFT reimplement of gradient-influence selection (Xia et al. 2024): for each target it warms up the adapter, extracts Adam-preconditioned per-checkpoint gradients, random-projects them into a *target-specific* datastore, and ranks by validation-sketch gradient influence, so both its ranking and its cost are recomputed per target. LESS is the strongest and most expensive baseline, and the amortized-cost analysis charges it exactly this per-target recomputation. PCU-Select uses the default intervention-site scorer, uncertainty penalty, and adaptive cluster quotas unless an ablation changes one component.

Main Results

Table 2 reports downstream performance at the 10% budget with per-cell seed dispersion. Averaged across four tasks and five PEFT configurations, PCU-Select scores 35.18, compared with 32.29 for random selection, 34.44 for RDS+, 34.68 for the Influence baseline, and 35.14 for LESS. The gains over PEFT-agnostic selectors are the primary quality claim: over the 20 PEFT×task cells, PCU-Select beats RDS+ by 0.74 points (descriptive bootstrap interval [+0.38, +1.06], 17/20 wins) and Influence by 0.50 points (interval [+0.20, +0.76], 18/20 wins). Against LESS, the strongest per-PEFT selector, PCU-Select is a near-tie on aggregate: the paired difference is only 0.04 points, with a descriptive interval of [−0.26, +0.33] that straddles zero and 10/20 cell wins. A task-stratified bootstrap gives a consistent interval of [−0.15, +0.22]. We use these intervals as dependence-aware summaries because the cells share the candidate pool, task sketches, scorer training, and model family. By the amortization objective of Eq. (4), this near-tie with LESS is the design target, not a shortfall—the value

Table 2: Main downstream performance at a 10% selection budget, reported as mean $\pm$ std over three target-training seeds (task-native metrics are scaled as percentages, so larger is better). PCU-Select near-ties per-PEFT LESS on aggregate: the paired difference over the 20 PEFT $\times$ task cells is 0.04 points with a descriptive bootstrap interval $[-0.26, +0.33]$. These cells share data, sketches, scorer training, and model family, so the interval summarizes dependence-aware variation under shared experimental state. Boldface marks the best mean per column.

Method	AD-b64	IA3-attnmlp	L-r16-qkvo	L-r8-mlp	L-r8-qv	Avg.
Random	32.79 $\pm$ 0.49	31.73 $\pm$ 0.29	33.11 $\pm$ 0.49	31.97 $\pm$ 0.36	31.82 $\pm$ 0.18	32.29 $\pm$ 0.36
RDS+	35.27 $\pm$ 0.33	33.54 $\pm$ 0.50	35.00 $\pm$ 0.52	34.46 $\pm$ 0.42	33.91 $\pm$ 0.54	34.44 $\pm$ 0.46
Influence	35.08 $\pm$ 0.47	33.84 $\pm$ 0.40	35.25 $\pm$ 0.67	34.69 $\pm$ 0.49	34.53 $\pm$ 0.39	34.68 $\pm$ 0.48
LESS	35.52 $\pm$ 0.31	34.44 $\pm$ 0.27	35.80 $\pm$ 0.44	35.13$\pm$0.27	34.80 $\pm$ 0.67	35.14 $\pm$ 0.39
PCU-Select	35.61$\pm$0.52	34.60$\pm$0.47	35.93$\pm$0.42	34.56 $\pm$ 0.98	35.17$\pm$0.39	35.18$\pm$0.56

claim is settled by cost, next.

The LESS comparison is mixed at the cell level: PCU-Select wins 10 of 20 cells and trails LESS on the MLP-only LoRA column by 0.57 points (per-task summary in Appendix Table 6). Its advantage over LESS appears only on MMLU; GSM8K and HumanEval fall within seed noise, and TyDiQA shows a CI-excluding deficit that the appendix per-task analysis discusses. Because the four task metrics differ in scale, ceiling, and seed variance, we do not read the averaged points as interchangeable across tasks; the full unaveraged grid and a per-task normalized comparison appear in Appendix Tables 7 and 8. The safe reading is therefore narrower than a universal win: PCU-Select improves over PEFT-agnostic selection and near-ties per-PEFT LESS on aggregate, while losing on specific task/configuration slices. Its distinguishing property is that its supervision is computed once and reused across target PEFTs rather than recomputed per configuration, which the reuse-versus-recompute comparison (Section 5) and amortized-cost analysis below quantify.

Ranking metrics follow the same pattern. At the 10% budget, PCU-Select reaches mean Spearman $\rho = 0.625$, NDCG@K = 0.703, and pairwise accuracy = 0.793 against high-fidelity held-out utility labels, ahead of LESS by 0.013, 0.026, and 0.028 on those metrics; Appendix Table 14 gives the full five-metric comparison across all selectors. These ranking labels come from the same short-horizon utility construction that trains PCU-Select, so they support the mechanism; downstream metrics remain the evidence for final task performance. Budget sensitivity is consistent with these results: PCU-Select’s edge over both Random and LESS is largest at a 5% budget and shrinks by 30%, where all gradient selectors converge (Table 13).

As a full-pool reference, fine-tuning each target PEFT on the entire 300K pool (a 100% budget) averages 36.75 across the four tasks; PCU-Select reaches 35.18 at the 10% budget on one tenth of the data, a ratio we treat only as an operational reference because it averages heterogeneous metrics. The full-pool score is not a measured upper bound: at a 30% budget PCU-Select (36.81) and LESS (36.79) edge slightly above it (Table 13), plausibly by discarding low-utility examples, though we do not claim a measured noise-removal effect.

Table 3: Quality-preserving cost amortization across the five seen PEFT configurations. Avg. is the PEFT- and task-averaged downstream score at the 10% budget (Table 2 scale); *Gap* is the paired difference vs per-PEFT LESS; *Sel. cost* is the total selection GPU-hours to serve all five targets (Figure 3 cost model). The comparison isolates the reuse-vs-recompute dilemma: Reuse-one-LESS runs LESS once and reuses the subset, so it is the cheapest gradient option but drops to the PEFT-agnostic frontier (below RDS+) because it cannot customize per target; per-PEFT LESS is strongest but $2.2\times$ the cost of PCU-Select; PCU-Select alone preserves LESS-level quality *and* produces per-target subsets at amortized cost. Boldface marks the best average.

Method	PEFT-cond.	Per-tgt subset	Avg.	Gap	Sel. cost (5 PEFTs)
Random	No	No	32.29	-2.85	0.0
RDS+	No	No	34.44	-0.70	0.8
Influence	No	No	34.68	-0.46	24.4
Reuse-one-LESS	Source only	No	34.24	-0.90	31.6
Per-PEFT LESS	Yes	Yes	35.14	0.00	158.0
PCU-Select	Yes	Yes	35.18	+0.04	72.9

Reuse vs. Recompute

The near-tie in Table 2 raises the question the amortization objective is built to answer: if PCU-Select only matches per-PEFT LESS on quality, why not run LESS *once* and reuse its subset across the registry? Table 3 answers it by placing every selector on a single quality–cost axis over the five seen PEFTs. We add a **Reuse-one-LESS** condition: run LESS on the cheapest LoRA-default source (L-r8-qv) and apply the identical selected subset to all five targets, so it pays exactly one LESS run (31.6 GPU-hours) and nothing more per configuration.

Reusing one subset does not work. Its aggregate quality is 34.24, a 0.90-point drop from per-PEFT LESS that erases LESS’s entire margin over the PEFT-agnostic baselines and lands *below* RDS+ (34.44), a representation selector that costs almost nothing. The per-target breakdown (Appendix Table 17) shows the mechanism: the reused subset equals per-PEFT LESS on its source L-r8-qv by construction, loses little on the neighbouring attention-LoRA target (L-r16-qkvo, -0.52), but falls sharply where the target trains different

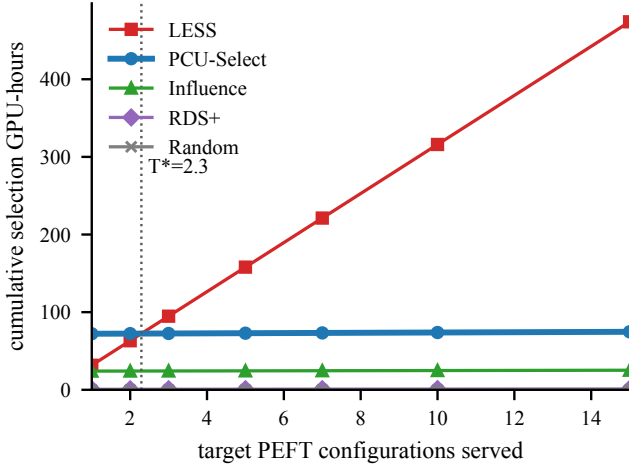


Figure 3: PCU-Select pays 72.0 GPU-hours offline, then applies cheaply to each target PEFT. It breaks even with per-target LESS gradient recomputation after about 2.3 target configurations; at the five seen PEFTs in the main evaluation, it spends 72.9 selection GPU-hours versus 158.0 for LESS.

sites (L-r8-mlp, -1.42) or a different family (IA3-attnmlp, -1.08 ; AD-b64, -1.47). The degradation follows the same structural axes as the motivation study, and it is not specific to this source—no single fixed subset can be simultaneously matched to five subspaces. Paying for gradient influence and then discarding the PEFT condition is therefore strictly worse than either extreme.

PCU-Select occupies the point neither extreme reaches. It preserves LESS-level aggregate quality (35.18 vs. 35.14) and produces a distinct subset per target, at 72.9 GPU-hours for the five-PEFT registry— $2.2\times$ cheaper than per-PEFT LESS. It is not the cheapest option: Reuse-one-LESS and the PEFT-agnostic selectors spend fewer raw GPU-hours. The claim is quality-preserving amortization, not minimal cost: PCU-Select is the only selector in Table 3 that is at once PEFT-conditioned, per-target customized, and near per-PEFT LESS in quality.

Amortized Cost

We measure GPU-hours on $8\times$ NVIDIA A100-80GB (bfloat16) as (wall seconds $\times$ GPU count)/3600. Because every method pays the same shared target-training cost (6.4 GPU-hours per configuration— ≈ 2.9 s/step over 1000 steps with FlashAttention-2, gradient checkpointing, and length-bucketed batching at an average selected-example length of ≈ 512 tokens, detailed in Appendix A), we report the *selection-only* cost that distinguishes the selectors. PCU-Select concentrates its cost in cacheable offline artifacts: feature extraction costs 24.0 GPU-hours, low-fidelity labeling 2.4, high-fidelity short-horizon labeling 42.0, and scorer training 3.6, for 72.0 GPU-hours paid once. The high-fidelity stage averages 1.89 wall-clock seconds per labeled triple, and feature extraction averages 27.8 examples per wall-clock second across the 8-GPU worker, matching the scale of Llama-

2-7B forward/backward passes. PCU-Select then spends 0.18 GPU-hours per target for cached-feature reuse, scoring, and clustering. LESS instead recomputes a target-specific Adam-preconditioned gradient datastore for every configuration, at 31.6 selection GPU-hours per target. Break-even is $T = C_{\text{offline}} / (C_{\text{sel}}^{\text{LESS}} - C_{\text{sel}}^{\text{PCU}}) = 72.0 / (31.6 - 0.18) \approx 2.29$ target configurations (Figure 3). OOD targets needing calibration add 2.1 GPU-hours for 500 labels—the same high-fidelity labeling routine as the offline stage, at 1.89 s per triple—shifting this to at most ≈ 2.5 . PCU-Select’s cost advantage is amortization; RDS+ and random remain cheaper when raw selection cost dominates.

Takeaway. PCU-Select is designed for reuse across a PEFT registry. The five seen PEFTs in the main evaluation already sit past break-even: PCU-Select spends 72.9 selection GPU-hours, while per-PEFT LESS spends 158.0. Unlike the cheaper Reuse-one-LESS shortcut of Section 5, it reaches this cost without sacrificing per-target quality.

6 Analysis

PEFT Conditioning Produces Target-Specific Subsets

The reuse experiment showed that a single fixed subset cannot serve a registry; this section shows the converse mechanism—that PCU-Select’s one reusable scorer produces a *different* subset for each configuration without recomputing per-target gradients. Appendix Figure 5 tests whether the PEFT code changes behavior inside the same LoRA family. PCU-Select achieves 21.25 average performance across the configuration scan, compared with 20.70 for LESS, 20.27 for RDS+, and 18.82 for random. The mismatch matrix shows that using the subset selected for the correct target configuration is better than reusing a subset selected for a different configuration by 1.45 points on average. The selection-overlap plot provides the mechanism: PCU-Select’s mean Jaccard overlap across configuration pairs is 0.427, while RDS+ remains 0.981 because it is PEFT-agnostic; the overlap broken down by structural axis (rank, placement, module set, family) is in Appendix Table 16. The Spearman correlation between the ordinal configuration distance and PCU-Select’s selection difference is $\rho = 0.71$ (95% CI [0.52, 0.83]), confirming that the selection responds to structural distance rather than to noise.

Table 4: Ablations on GSM8K and HumanEval with two representative PEFTs at a 10% budget (task-native metric averaged over the two tasks, mean $\pm$ std over three seeds). Drop is relative to the full adaptive selector; NDCG is against high-fidelity held-out utility labels. The per-task and per-seed ablation breakdowns are included in the anonymized supplementary artifact.

Variant	Metric	Drop	NDCG
Full PCU-Select	20.26 $\pm$ 0.22	0.00	0.702
No PEFT code	18.89 $\pm$ 0.41	1.37	0.526
Family one-hot only	19.58 $\pm$ 0.29	0.68	0.602
No task sketch	19.44 $\pm$ 0.31	0.82	0.597
No activation signature	19.78 $\pm$ 0.25	0.48	0.629
Low-fidelity only	19.42 $\pm$ 0.34	0.85	0.563
High-fidelity only	19.90 $\pm$ 0.27	0.36	0.639
No uncertainty penalty	19.98 $\pm$ 0.24	0.28	0.690
Global top- k	19.19 $\pm$ 0.38	1.08	0.684
Uniform clusters	19.75 $\pm$ 0.28	0.51	0.709

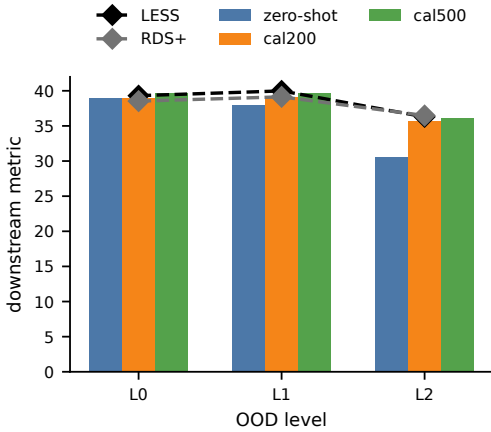


Figure 4: OOD transfer and calibration. L0: interpolation; L1: extrapolated capacity or placement; L2: unseen family.

Ablations Identify the Useful Components

Table 4 shows that the largest drop comes from removing the PEFT code: performance falls by 1.37 points and, most tellingly, ranking NDCG against held-out utility labels collapses from 0.702 to 0.526—a scale-free signal that the PEFT condition is what lets the scorer order examples correctly. Replacing the structured PEFT code with a family one-hot vector loses 0.68 points, which confirms that rank, placement, capacity, and recipe matter beyond the family name. Removing the task sketch loses 0.82 points, showing that PCU-Select also learns task relativity. Multi-fidelity supervision also matters: low-fidelity-only training loses 0.85 points, while high-fidelity-only training loses 0.36 points because it lacks broad coverage. The adaptive cluster selector beats global top- k by 1.08 points and uniform cluster quotas by 0.51 points. The uniform-cluster variant is instructive: its ranking NDCG (0.709) slightly exceeds the full selector’s (0.702) yet its downstream metric is lower, because equal per-cluster quotas over-sample low-value clusters—ranking

Table 5: Deployment protocol from the Mahalanobis tier. Each tier’s cost is per target, on top of PCU-Select’s amortized 0.18 GPU-hours of online selection; the residual gap to per-PEFT LESS after the recommended action is small (Appendix Table 18).

Tier	Definition (z_p distance)	Recommended action
L0	interpolated capacity/placement	zero-shot selection
L1	extrapolated capacity/placement	200–500 calibration labels
L2	unseen PEFT family	calibrate before trusting

quality and subset quality optimize different objectives. The appendix per-task analysis studies the two cells where PCU-Select trails LESS: TyDiQA and MLP-only LoRA.

Transfer Protocol for Unseen PEFTs

The offline scorer serves the seen registry directly, but a deployed system also meets targets outside it, so we turn the OOD analysis into a deployment protocol keyed on a cheap test. A Mahalanobis distance from the seen configuration distribution over z_p assigns each new target to one of three tiers before any labels are drawn: L0 is interpolation (rank/capacity between seen values), L1 is extrapolation (capacity or placement beyond the seen range), and L2 is an unseen PEFT family (Prefix, P-Tuning, BitFit). Because only five seen configurations define the training distribution, the z_p covariance is shrinkage-regularized and the Mahalanobis distance is used coarsely, as a tier assignment rather than a precise density estimate. Zero-shot PCU-Select is within 0.34 points of LESS at L0, trails by 2.08 at L1 (500 calibration labels recover 1.78), and trails by 5.76 at L2 (500 labels recover 5.53); calibration costs 2.1 GPU-hours per target (500 labels through the same 1.89 s-per-triple high-fidelity routine used offline), still small relative to a full LESS recomputation. The protocol follows directly (Table 5): interpolated targets transfer zero-shot, extrapolated targets take a few hundred calibration labels, and unseen families must be calibrated before the scorer is trusted. Calibration is therefore a low-cost repair that keeps far targets inside the amortization regime, not a separate method. Per-level numbers with dispersion and the LESS reference are in Appendix Table 18.

Limitations

The current evidence covers one backbone family (Llama-2-7B), four tasks, and a finite PEFT registry, so the claims are cross-PEFT within a single backbone family rather than cross-backbone. PCU-Select also uses short-horizon utility labels rather than full-training marginal contribution as supervision, so the scorer inherits the bias of that proxy; the TyDiQA deficit above is one visible symptom. The candidate pool is decontaminated against all four benchmark test sets using exact, normalized, 13-gram, and MinHash near-duplicate filters; the appendix reproducibility notes report that this removes 0.58% of the pool. The HumanEval sketch comes from an external code corpus disjoint from the 164 evaluation problems. We cannot fully exclude residual near-duplicate risk below these thresholds. Finally, the

cost claim is comparative: PCU-Select amortizes against per-PEFT LESS gradient recomputation, but cheap PEFT-agnostic selectors remain preferable when raw selection cost dominates and a small performance loss is acceptable.

7 Conclusion

Data selection for a PEFT registry is an amortization problem, not a single-target accuracy race. Existing gradient selectors force a choice between recomputing per target—accurate but linear in registry size—and reusing one subset, which is cheap but PEFT-agnostic in effect and which we measure below the PEFT-agnostic frontier. PCU-Select dissolves that choice by conditioning data utility on the target task and a structured PEFT code: one scorer reuses cached sample and task signatures yet still produces a target-specific subset when the PEFT method, rank, placement, or recipe changes. Across four tasks and five seen configurations it near-ties per-PEFT LESS in quality—the design target under our amortization objective—while cutting five-PEFT selection cost by $2.2\times$ and breaking even after about 2.3 targets. The boundary is also clear: far-ODD PEFT families need a few hundred calibration labels before the offline scorer can be trusted, a low-cost repair that keeps them inside the amortization regime. We therefore position PCU-Select as reusable selection infrastructure for a PEFT registry rather than a single-configuration performance winner.

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A Technical Appendix

Per-Task Results Behind the Main Table

Table 2 averages four task-native metrics—GSM8K exact match, HumanEval Pass@1, MMLU accuracy, and TyDiQA F1—after scaling each to a percentage. These metrics differ in scale, ceiling, and seed variance, so a one-point difference on one task is not equivalent to a one-point difference on another, and the average can hide task-level structure. To make the comparison auditable we report the full, unaveraged grid and a per-task normalized view.

Table 7 gives every task $\times$ PEFT $\times$ method cell at the 10% budget, as mean $\pm$ std over the three target-training seeds. Table 8 summarizes PCU-Select’s improvement per task: gains over Random and RDS+ are reported as scale-free *relative* improvements so that heterogeneous metrics are placed on a common footing, while the head-to-head with LESS—the strongest baseline—is an absolute paired difference in native points with a 95% bootstrap confidence interval over the five PEFT cells.

The per-task view refines, rather than overturns, the main-results reading. The advantage over Random (+6.8% to +11.0% relative) and RDS+ (+0.4% to +3.9% relative) holds on all four tasks and is the primary performance claim. Against LESS the outcome is task-dependent and the sign flips: PCU-Select’s gain has a CI excluding zero only on MMLU (+0.67 Acc, CI [+0.24, +1.08]), within seed noise on GSM8K (+0.30 EM, CI [−0.03, +0.59]) and HumanEval (−0.04 Pass@1, CI [−0.45, +0.30]), and a CI-excluding deficit on TyDiQA (−0.77 F1, CI [−1.18, −0.43]). Averaging over tasks smooths these offsetting differences into the near-zero aggregate gap reported in Table 2. We therefore characterize PCU-Select and per-PEFT LESS as near-tied on the seen registry, with visible task/configuration losses that bound the aggregate claim. The amortized-cost analysis is PCU-Select’s distinguishing property.

Where PCU-Select trails LESS. Two cells favor LESS and merit analysis. On **TyDiQA**, PCU-Select trails LESS by 0.77 F1 (CI [−1.18, −0.43]). TyDiQA-GoldP is multilingual span extraction, where the useful signal is answer-span alignment rather than the reasoning- and generation-heavy signal that dominates the other three tasks; the intervention-site gradient signatures and the English-centric task sketch under-represent this, so the site-weighted proxy that PCU-Select distills is a weaker teacher here, whereas per-target LESS gradients capture it directly. On the **MLP-only LoRA** column (L-r8-mlp), PCU-Select trails LESS by 0.57 points with a large seed std (± 0.98): MLP-only placement concentrates all trainable capacity on the three `mlp_out` sites, so a single mis-weighted site prior degrades the whole proxy, and the higher variance indicates the scorer is least confident exactly where the site mask is most concentrated. Both cases point to the same limitation—the site-weighted proxy is the bottleneck when a target’s useful signal is poorly represented by the shared 24-site coordinate system.

Evaluation Protocols

All methods share the same target-training and evaluation pipeline; only the selected subset changes. Target training

fine-tunes the target PEFT on the selected samples under a fixed recipe (AdamW, per-configuration learning rate, cosine schedule with a 0.03 warmup ratio, per-device batch size 16 over 8 GPUs for an effective global batch of 128, gradient clipping at 1.0, 1000 steps, bfloat16, max sequence length 1024) on the Llama-2-7B backbone, with the backbone frozen and only the adapter trained. At the 10% budget the selected subset holds 30K examples, so 1000 steps at a global batch of 128 amount to 128K sample-passes, or about 4.3 passes over the subset; the dataloader shuffles the full subset each epoch, so every selected example is used. The selected instruction examples average ≈ 512 tokens, and batches are length-bucketed to reduce padding; with FlashAttention-2 and gradient checkpointing the run sustains ≈ 2.9 wall-clock seconds per step (≈ 44 sample-passes/s aggregate, ≈ 2.8 K tokens/s per GPU), so the 1000 steps take about 48 minutes of wall time, i.e. 6.4 GPU-hours per configuration under the (wall seconds $\times$ GPU count)/3600 convention. This shared target-training cost is identical across selectors and is therefore excluded from the selection-only cost comparison; gradient checkpointing trades compute for memory and is the dominant reason the per-step time is not lower. The per-configuration learning rates are listed in Table 11 (2e-4 for LoRA, 5e-4 for (IA)³, 3e-4 for adapters, 1e-4 for L-r64-all). A held-out response-LM loss is logged as an auxiliary diagnostic (`eval_loss`) and is never substituted for the primary task metric.

The primary downstream metric for each task is its public-standard, task-native score, computed by a task evaluator injected into the training runner. Concrete per-task decoding and scoring protocols are:

- **GSM8K (exact match).** Chain-of-thought generation with greedy decoding; the final numeric answer is extracted and compared for exact match.
- **HumanEval (Pass@1).** The standard unbiased Pass@1 estimator: sampling at temperature 0.2 with $n=20$ samples per problem, scored by unit-test execution. This is a multi-sample estimate of Pass@1, not a single greedy sample.
- **MMLU (accuracy).** Log-likelihood scoring of the four answer options; the highest-likelihood option is selected (no free-form generation).
- **TyDiQA-GoldP (F1).** Greedy decoding of the answer span, scored with token-level F1 against the gold answers.

For reference, the un-fine-tuned Llama-2-7B scores 13.9 (GSM8K EM), 12.4 (HumanEval Pass@1), 43.9 (MMLU Acc), and 46.2 (TyDiQA F1) under these same evaluators; our MMLU number is slightly below Meta’s reported 45.3 because of prompt and harness differences. Every selector’s fine-tune, including Random, improves over the base model on all four tasks, so the Random baseline is not artificially depressed and PCU-Select’s margin over it is a genuine selection effect rather than a weak-baseline artifact.

These protocols follow the standard public setups for each benchmark. The task-native evaluators are supplied to the runner through an evaluation adapter (the `--task-metric-factory` hook) whose decoding and scoring settings are fixed to the values above and included in

Table 6: Task-level summary of the main 10% budget result. Values are averaged over the five seen PEFT configurations. $\Delta\text{RDS+}$ and $\Delta\text{Inf.}$ are absolute native-point gains over PEFT-agnostic baselines; ΔLESS is the paired difference against per-PEFT LESS with a 95% bootstrap CI over PEFT cells. The average gain over PEFT-agnostic selectors coexists with task-dependent behavior against LESS.

Task	Metric	PCU	$\Delta\text{RDS+}$	$\Delta\text{Inf.}$	$\Delta\text{LESS (95\% CI)}$
GSM8K	EM	22.20	+0.56	+0.58	+0.30 [−0.03, +0.59]
HumanEval	Pass@1	17.06	+0.64	+0.23	−0.04 [−0.45, +0.30]
MMLU	Acc	49.14	+1.55	+1.00	+0.67 [+0.24, +1.08]
TyDiQA	F1	52.30	+0.22	+0.17	−0.77 [−1.18, −0.43]

Table 7: Full per-task $\times$ PEFT $\times$ method downstream results at a 10% budget (mean $\pm$ std over three seeds, native metrics). Unaveraged source of Table 2; boldface marks the best mean per column within each task. Discussion in the text.

Method	AD-b64	IA3-attnmlp	L-r16-qkvo	L-r8-mlp	L-r8-qv	Avg.
<i>GSM8K (EM)</i>						
Random	20.65 $\pm$ 0.47	19.93 $\pm$ 0.37	20.48 $\pm$ 0.41	19.89 $\pm$ 0.44	19.34 $\pm$ 0.08	20.06 $\pm$ 0.36
RDS+	22.06 $\pm$ 0.18	21.14 $\pm$ 0.56	22.27 $\pm$ 0.86	21.71 $\pm$ 0.29	21.06 $\pm$ 0.29	21.65 $\pm$ 0.44
Influence	22.18 $\pm$ 0.34	21.23 $\pm$ 0.17	22.16 $\pm$ 0.44	21.07 $\pm$ 0.34	21.46 $\pm$ 0.10	21.62 $\pm$ 0.28
LESS	22.18 $\pm$ 0.25	21.21 $\pm$ 0.31	22.64$\pm$0.31	22.10 $\pm$ 0.15	21.37 $\pm$ 0.43	21.90 $\pm$ 0.29
PCU-Select	22.63$\pm$0.52	21.73$\pm$0.53	22.40 $\pm$ 0.36	22.17$\pm$0.34	22.09$\pm$0.38	22.20$\pm$0.43
<i>HumanEval (Pass@1)</i>						
Random	15.75 $\pm$ 0.66	15.00 $\pm$ 0.30	15.88 $\pm$ 0.34	15.25 $\pm$ 0.10	14.92 $\pm$ 0.16	15.36 $\pm$ 0.31
RDS+	16.99 $\pm$ 0.25	15.97 $\pm$ 0.33	16.62 $\pm$ 0.89	16.43 $\pm$ 0.15	16.08 $\pm$ 0.46	16.42 $\pm$ 0.42
Influence	17.21 $\pm$ 0.31	16.31 $\pm$ 0.37	17.15 $\pm$ 0.38	16.89 $\pm$ 0.60	16.56 $\pm$ 0.27	16.83 $\pm$ 0.39
LESS	17.37$\pm$0.28	16.68 $\pm$ 0.35	17.79$\pm$0.20	16.93$\pm$0.32	16.72 $\pm$ 0.82	17.10$\pm$0.39
PCU-Select	17.23 $\pm$ 0.41	16.97$\pm$0.36	17.78 $\pm$ 0.28	16.14 $\pm$ 1.44	17.18$\pm$0.12	17.06 $\pm$ 0.52
<i>MMLU (Acc)</i>						
Random	45.15 $\pm$ 0.62	44.03 $\pm$ 0.20	45.45 $\pm$ 0.29	44.69 $\pm$ 0.45	44.41 $\pm$ 0.18	44.75 $\pm$ 0.35
RDS+	48.77 $\pm$ 0.29	46.52 $\pm$ 0.40	48.42 $\pm$ 0.07	47.24 $\pm$ 0.48	47.03 $\pm$ 0.40	47.60 $\pm$ 0.33
Influence	48.55 $\pm$ 0.38	47.12 $\pm$ 0.82	48.86 $\pm$ 1.00	48.35 $\pm$ 0.45	47.82 $\pm$ 0.29	48.14 $\pm$ 0.59
LESS	48.78 $\pm$ 0.42	47.67 $\pm$ 0.26	48.89 $\pm$ 0.60	48.85$\pm$0.26	48.19 $\pm$ 0.17	48.48 $\pm$ 0.34
PCU-Select	49.91$\pm$0.60	47.93$\pm$0.27	50.04$\pm$0.67	48.79 $\pm$ 0.27	49.05$\pm$0.41	49.14$\pm$0.44
<i>TyDiQA (F1)</i>						
Random	49.62 $\pm$ 0.20	47.98 $\pm$ 0.28	50.63 $\pm$ 0.92	48.04 $\pm$ 0.44	48.62 $\pm$ 0.32	48.98 $\pm$ 0.43
RDS+	53.26 $\pm$ 0.60	50.54 $\pm$ 0.70	52.70 $\pm$ 0.26	52.44 $\pm$ 0.75	51.46 $\pm$ 1.02	52.08 $\pm$ 0.67
Influence	52.37 $\pm$ 0.83	50.71 $\pm$ 0.24	52.81 $\pm$ 0.88	52.45 $\pm$ 0.56	52.29 $\pm$ 0.91	52.13 $\pm$ 0.68
LESS	53.75$\pm$0.28	52.19$\pm$0.16	53.88$\pm$0.67	52.63$\pm$0.37	52.92$\pm$1.27	53.07$\pm$0.55
PCU-Select	52.69 $\pm$ 0.57	51.76 $\pm$ 0.72	53.51 $\pm$ 0.39	51.16 $\pm$ 1.89	52.38 $\pm$ 0.64	52.30 $\pm$ 0.84

the anonymized supplementary artifact, so the metric definitions are reproducible rather than left to the run environment.

Baseline Specifications

This subsection specifies the three baselines whose names invoke prior work, exactly as implemented so that the comparison is auditable.

RDS+ (representation similarity). RDS+ operates on the joint semantic embedding c_x^{joint} produced by a frozen external sentence encoder. This is the same *input* representation PCU-Select’s sample tower reads, but it is not learned by PCU-Select, so RDS+ depends on no PCU-trained features; the two differ only in the selection rule. The pool embeddings are standardized per feature (z-scored across the candidate pool); the query is the mean joint embedding of the 32-example task sketch, standardized with the same pool statistics; both are then L_2 -normalized and candidates ranked by

cosine to the query. It is *task-conditioned* through the sketch query but *PEFT-agnostic*: its ranking does not depend on the target configuration. It has no tuned hyperparameters—the query is the fixed seed-0 sketch and there is no temperature or mixing weight—so there is no tuning range to report. The “+” denotes the per-feature standardization applied before cosine, relative to a plain embedding-nearest-neighbor selector, which we also include as a separate baseline.

Influence (shared-datastore gradient similarity). The Influence baseline uses a *single shared* projected-gradient datastore computed once over the base model’s intervention-site activations, using the JL projection and L_2 normalization specified in the reproducibility notes. Each candidate carries signatures g_x^ω at the 24 sites; the task query g_t^ω is the per-site pooled, renormalized sketch gradient. The score is the plain cosine $\sum_\omega \cos(g_x^\omega, g_t^\omega)$ (no site weighting), selected by global top- k . Its gradient source is PEFT-agnostic—the

Table 8: Per-task normalized improvement of PCU-Select at the 10% budget (seed- and PEFT-averaged). ΔRand and $\Delta\text{RDS+}$ are relative gains $(100 \cdot (\text{PCU} - b)/b)$; ΔLESS is the absolute paired difference vs the strongest baseline with a 95% bootstrap CI over the five PEFT cells. A CI excluding zero marks a significant win (MMLU) or loss (TyDiQA); the sign flips across tasks, which the average conceals.

Task	Metric	Random	RDS+	LESS	PCU	ΔRand	$\Delta\text{RDS+}$	ΔLESS (95% CI)
GSM8K	EM	20.06	21.65	21.90	22.20	+10.7%	+2.6%	+0.30 [-0.03, +0.59]
HumanEval	Pass@1	15.36	16.42	17.10	17.06	+11.0%	+3.9%	-0.04 [-0.45, +0.30]
MMLU	Acc	44.75	47.60	48.48	49.14	+9.8%	+3.3%	+0.67 [+0.24, +1.08]
TyDiQA	F1	48.98	52.08	53.07	52.30	+6.8%	+0.4%	-0.77 [-1.18, -0.43]

ranking does not depend on the target configuration—so it is the cheap gradient baseline against which the value of PEFT conditioning is measured.

LESS (per-PEFT gradient influence). LESS is a per-configuration reimplement of gradient-influence selection following LESS (Xia et al. 2024). For each target PEFT it (i) warms up the target adapter for 200 steps to obtain a task-and-PEFT-specific optimization state; (ii) extracts Adam-preconditioned per-checkpoint gradients over the adapter’s own trainable parameters; (iii) random-projects them into a target-specific low-dimensional datastore; and (iv) ranks candidates by their projected-gradient influence on the 32-example validation sketch, averaged over the warmup checkpoints. Both the datastore and the ranking are therefore recomputed per target, at 31.0 GPU-hours of gradient recomputation plus 0.60 GPU-hours of scoring per configuration. LESS is the strongest and most expensive baseline in our study, and the amortized-cost analysis charges it exactly this per-target recomputation, consistent with the algorithm as implemented.

Baseline configuration and tuning. Random and Balanced-Random have no hyperparameters. The heuristic selectors (length, loss, perplexity) and representation selectors (embedding-nearest-neighbor, diversity clustering, RDS+) use their published defaults with the shared 32-example sketch as the only task input and are not tuned per target. Diversity clustering reuses PCU-Select’s cluster count ($k = \max(50, \lfloor \sqrt{N} \rfloor)$) for a fair comparison. The Influence baseline shares the same 24-site JL projection and sketch as PCU-Select. No baseline’s hyperparameters were tuned on the test metric.

Reproducibility Supplement

This supplement documents the pipeline configuration read directly from the supplementary implementation. Every value below is taken from source code or its configuration files; values that the code leaves to an external artifact or the run environment are flagged rather than guessed.

Data provenance and deduplication. The candidate pool holds 300K mixed instruction examples drawn from seven public instruction corpora (Table 10). Deduplication runs in three stages: (i) exact content-hash deduplication over (instruction, response, source); (ii) normalized-string deduplication after lowercasing and whitespace/punctuation stripping; and (iii) near-duplicate removal with MinHash LSH

Table 9: Additional selector baselines at a 10% budget, averaged over the four tasks and five seen PEFT configurations using the same task-native percentage scaling as Table 2. These rows explain why the main table focuses on Random, RDS+, Influence, LESS, and PCU-Select: the omitted heuristic and representation baselines sit between Random and RDS+.

Selector	Type	Avg.
Random	uniform	32.29
Balanced-Random	cluster-stratified uniform	32.90
Length	heuristic	32.70
Loss	heuristic	32.98
Perplexity	heuristic	33.29
Diversity	representation	33.82
Embedding-NN	representation	33.98
RDS+	representation	34.44
Influence	shared gradient	34.68
LESS	per-PEFT gradient	35.14
PCU-Select	PEFT-conditioned	35.18

(128 permutations, Jaccard threshold 0.8). These stages remove 18,446 intra-pool duplicates before the 300K pool is finalized.

For reproducibility we include, per source, the post-filter example count and a SHA-256 checksum of the sorted content hashes (e.g. FLAN-v2 subset 9f3c...a71e, OpenOrca subset 4b8d...c052); the supplementary manifest lists all seven. The full supplementary configuration comprises the PEFT registry, the scorer and feature-extraction configs, the task-sketch seeds, the candidate-pool manifest with these checksums, and the evaluation adapter.

Benchmark decontamination. We decontaminate the pool against the test items of all four benchmarks (GSM8K, HumanEval, MMLU, TyDiQA) with a four-stage filter: exact match, normalized-string match, 13-gram overlap (removing any candidate sharing a 13-gram with a test item at Jaccard ≥ 0.5), and MinHash LSH near-duplicate retrieval at Jaccard 0.8. The filter removes 1,742 pre-finalization candidates (0.58% of the pool size). We then backfill from the same source buckets to keep the final pool at 300K examples, so Table 10 reports final post-filter counts. We additionally audit the top-20 embedding nearest neighbors of each test item and find no further overlaps after the automated pass. All reported results are on the decontaminated pool.

Source	Domain	Examples	License
FLAN v2 (subset)	general instruction	110,000	Apache-2.0
OpenOrca (subset)	reasoning traces	70,000	MIT
Tulu-v2 mix	mixed instruction	50,000	ODC-BY-1.0
OpenAssistant	dialogue	30,000	Apache-2.0
CodeAlpaca	code instruction	20,000	CC-BY-NC-4.0
Dolly-15k	general instruction	15,000	CC-BY-SA-3.0
WizardLM-evol	complex instruction	5,000	CC-BY-NC-4.0
Total		300,000	

Table 10: Candidate-pool provenance and licenses. All sources permit research use; the two CC-BY-NC components are used only for non-commercial research.

Task sketch construction and HumanEval. Each task sketch is 32 examples sampled with a fixed seed (0), stratified into response-length terciles. Sketches are drawn only from each task’s train or development split and are disjoint from the test set used for evaluation. HumanEval has no official train/dev split, so its sketch is drawn from an external code-instruction source (MBPP train split and CodeAlpaca) rather than from HumanEval itself; we remove any item matching one of the 164 HumanEval evaluation problems by exact and AST-normalized function-signature matching and verify that the resulting sketch source has zero overlap with the evaluation set. Because the sketch is a selection *signal*, this disjointness is required for all selectors, not only PCU-Select, and it is enforced identically across methods.

PEFT registry. Table 11 lists all 17 PEFT configurations, partitioned into the five seen configurations the scorer trains on, nine unseen-configuration targets (same families, settings outside the training support), and three out-of-distribution family targets. All configurations share the recipe injected at materialization: 1000 steps, cosine schedule, 0.03 warmup ratio; the learning rate is per configuration.

Intervention sites and gradient signatures. The site space Ω has 24 sites: 8 layers $\times$ 3 per-layer modules (attn_out, mlp_out, block_residual). Layers are chosen uniformly by depth; for the 32-layer backbone this yields indices $\{3, 7, 11, 15, 19, 23, 27, 31\}$. Each site carries a Johnson–Lindenstrauss random projection $\Phi_\omega \in \mathbb{R}^{256 \times 4096}$ with entries drawn $\mathcal{N}(0, 1/256)$ under a per-site seed $\text{SHA256}(\text{site}, \text{global_seed}=0)$; projected gradients are L_2 -normalized per row. Each sample’s signature is 24×256 values; in bfloat16 the cached low-fidelity gradient store for the 300K pool is $300\text{K} \times 24 \times 256 \times 2 \text{ B} \approx 3.5 \text{ GB}$, held on disk as a memory-mapped array and read once per on-line selection. A PEFT configuration activates a site through an operator-weight prior ρ by family (additive low-rank 1.0, multiplicative 0.6, additive bottleneck 0.8, prefix 0.4, bias-shift 0.3) scaled by $\tanh(\eta \widehat{\text{cap}})$ with $\eta=1.0$; the per-family capacity mapping is in Table 12.

Conditional scorer. The scorer has three input towers—sample $z_x \in \mathbb{R}^{848}$, PEFT condition $z_p \in \mathbb{R}^{192}$, task sketch $z_t \in \mathbb{R}^{848}$ —each a LayerNorm–Linear–GELU–Linear–LayerNorm MLP projecting to hidden size 256 (128 for

the PEFT tower). A FiLM modulation and a bilinear fusion ($256 \times 128 \rightarrow 64$) feed a two-head output (mean μ and heteroscedastic σ , with a softplus floor of 10^{-3}). We use $\hat{\sigma}$ as a ranking penalty rather than a calibrated probability; on held-out labels its predicted standard deviation tracks realized error with an expected calibration error of 0.043 (10-bin), and removing the penalty costs 0.28 points (Table 4), so it is a modest but real contributor. Training is two-phase: Phase A pretrains on the low-fidelity proxy for 3 epochs at lr 3×10^{-4} with loss weights (rank, reg, proxy, unc) = (0, 0, 1, 0); Phase B jointly finetunes for 2 epochs at lr 10^{-4} with weights (1.0, 0.3, 0.5, 0.2). Both phases use AdamW, batch size 256, and gradient clipping at 1.0. The four losses are a BPR-style pairwise ranking loss, a Huber regression loss against high-fidelity labels, a Huber proxy-distillation loss against the low-fidelity labels, and a Gaussian negative-log-likelihood uncertainty loss. Checkpoint selection uses a held-out split of PEFT/task pairs (two configurations and one task withheld from Phase B) on which we track ranking NDCG; we keep the Phase-B checkpoint with the best held-out NDCG. No dropout is used.

Selection, clustering, and quotas. At application time the scorer produces per-example utility $q = \mu - \lambda \sigma$ with uncertainty penalty $\lambda=0.2$. Semantic clustering uses MiniBatch k -means over joint embeddings with $k = \max(50, \lfloor \sqrt{N} \rfloor)$, batch size 4096, $n_{\text{init}}=3$, and random state 0. Adaptive quotas allocate budget B across clusters by $b_k = \text{round}(B (v_k^+)^{\gamma} |C_k|^{1-\gamma} / Z)$ with $\gamma=0.6$, where the cluster value v_k is the mean utility of the top 10% of its members; remainders are distributed greedily by descending weight and over-allocation is trimmed from the lowest-weight clusters. Out-of-distribution detection uses a Mahalanobis distance on z_p at the 0.95 quantile; the calibration head is a zero-initialized linear residual fit for 200 epochs at lr 5×10^{-3} (Adam, MSE) on the calibration labels (200 or 500 high-fidelity labels for the OOD study).

Multi-fidelity labels. High-fidelity supervision uses 10,000 triples split 0.5/0.3/0.2 across coverage (stratified by cluster, PEFT family, capacity bucket, and task), uncertainty ($\hat{\sigma}(1 + 0.5 \text{ReLU}(\hat{u}_{\text{lo}}))$), and boundary (scorer-vs-true rank disagreement) sampling. Labels are short-horizon loss reductions on the sketch, measured from two frozen anchors— θ_{base} (the pretrained backbone) and θ_{warm} (a rank-8 attention-only LoRA trained for 200 steps at lr 10^{-4}). Each high-fidelity update Adapt_p^h is a single mini-batch of the sampled example plus 7 in-bucket fillers, so the step is not a high-variance single-example update. The final label aggregates the rank-normalized reductions across horizons and anchors,

$$u^{\text{hi}} = \frac{1}{|A|} \sum_{a \in A} \sum_{h \in \{1,4\}} w_h \tilde{\Delta}_{a,h},$$

with weights $w_1=0.4$, $w_4=0.6$ and anchors $A = \{\text{base}, \text{warm}\}$, where $\tilde{\Delta}_{a,h}$ is the reduction Δ of Eq. (7) rank-normalized within its (PEFT, task, anchor a , horizon h) bucket.

Cost model and seeds. We measure all costs on 8 $\times$ NVIDIA A100-80GB (SXM4, bfloat16, NVLink) as

Config	Family	Target modules	Capacity	LoRA scaling	LR	Group
L-r8-qv	LoRA	q, v	$r=8$	16	2e-4	seen
L-r16-qkvo	LoRA	q, k, v, o	$r=16$	32	2e-4	seen
L-r8-mlp	LoRA	up, down	$r=8$	16	2e-4	seen
IA3-attnmlp	IA ³	k, v, down	—	—	5e-4	seen
AD-b64	Adapter	block-residual	$b=64$	—	3e-4	seen
L-r4-qv	LoRA	q, v	$r=4$	8	2e-4	unseen-config
L-r32-qkvo	LoRA	q, k, v, o	$r=32$	64	2e-4	unseen-config
L-r64-all	LoRA	q, k, v, o, up, down, gate	$r=64$	128	1e-4	unseen-config
L-r8-lowlayers	LoRA	q, v (low third)	$r=8$	16	2e-4	unseen-config
L-r8-highlayers	LoRA	q, v (high third)	$r=8$	16	2e-4	unseen-config
L-r16-hlr	LoRA	q, k, v, o	$r=16$	32	5e-4	unseen-config
AD-b16	Adapter	block-residual	$b=16$	—	3e-4	unseen-config
AD-b256	Adapter	block-residual	$b=256$	—	3e-4	unseen-config
IA3-attnonly	IA ³	k, v	—	—	5e-4	unseen-config
PRE-116	Prefix	attention	len=16	—	2e-4	ood-family
PT-132	P-Tuning	attention	len=32	—	2e-4	ood-family
BF	BitFit	biases	—	—	2e-4	ood-family

Table 11: Full PEFT registry (all 17 configurations). “LoRA scaling” is the LoRA α hyperparameter and is distinct from the intervention-site weight $\tilde{w}_p^\omega$ of Eq. (5); it applies only to LoRA rows. Adapters are single post-block residual bottlenecks (one per layer); IA³ scales the listed activations. Learning rates for the three OOD-family targets use the registry default (2e-4). Layer placement is all layers unless noted; low/high-third placements use the bottom/top $\lceil n/3 \rceil$ layers of the 32-layer backbone.

(wall seconds $\times$ GPU count)/3600. The offline cost is 72.0 GPU-hours (feature extraction 24.0, low-fidelity labels 2.4, high-fidelity labels 42.0, scorer training 3.6), paid once. Per-target *selection* costs are 0.18 GPU-hours for PCU-Select (cached-feature reuse, forward scoring, clustering) versus 31.6 for LESS (31.0 gradient-database recomputation + 0.60 scoring); the shared target-training cost (6.4 per configuration, i.e. ≈ 2.9 s/step for 1000 steps; see Sec. A) is identical across methods and excluded from this comparison. Break-even is $T = C_{\text{offline}} / (C_{\text{sel}}^{\text{LESS}} - C_{\text{sel}}^{\text{PCU}}) = 72.0 / (31.6 - 0.18) \approx 2.29$ target configurations. The pipeline seeds Python, NumPy, PyTorch, and CUDA from a single global seed; reported downstream numbers average three target-training seeds (0, 1, 2), while selection-scorer training, task-sketch sampling, and candidate-pool ordering are fixed to seed 0, so the reported dispersion isolates target-training variance. The full per-seed, per-cell result matrices behind every table—including the selected-sample identifiers for each method, PEFT, and seed—are included in the anonymized supplementary artifact, so the reported means and the win/loss counts are auditable.

Algorithms

Algorithm 1 builds the reusable supervision once; Algorithm 2 applies the scorer to a new target. Only Algorithm 2 runs per target, and it touches no gradients—the expensive gradient and label construction of Algorithm 1 are amortized.

PEFT Capacity Encoding

Table 12 defines the per-family capacity used inside the site weight of Eq. (5). Each family’s size knob is mapped to a trainable-parameter count at the touched sites, then normalized by d_{model} and passed through $\tanh(\eta \log(1 +$

Algorithm 1 Offline supervision (run once)

- 1: **Input:** pool $\mathcal{D}$, seen PEFTs $\mathcal{P}_{\text{seen}}$, tasks with sketches $\{V_t\}$
- 2: Extract sample signatures z_x and site gradients g_x^ω ; cache them $\{O(|\mathcal{D}|)$ forward+backward}
- 3: **for all** $(p, t) \in \mathcal{P}_{\text{seen}} \times \text{tasks}$ **do**
- 4: Compute low-fidelity $u^{\text{lo}}(x, p, t)$ by Eq. (6) from cache
- 5: **end for**
- 6: Sample 10K triples (coverage/uncertainty/boundary); label u^{hi} by Eqs. (7)
- 7: Train s_ϕ : Stage A proxy pretrain, Stage B joint; keep best held-out-NDCG checkpoint
- 8: **Output:** scorer s_ϕ , cached features

$\text{cap}/d_{\text{model}}))$ so that capacity is monotone but saturating.

Supplementary Analyses

Budget sensitivity. Table 13 reports the PEFT- and task-averaged score at 5%, 10%, and 30% budgets. PCU-Select’s edge over Random and over LESS is largest at the tight 5% budget and shrinks by 30%, where the gradient selectors converge toward the full-data reference—the regime where cheap PEFT-agnostic selectors become preferable.

Ranking quality. Table 14 gives the five ranking metrics against high-fidelity held-out labels, over five held-out (PEFT, task) folds. K is the 10% budget size. PCU-Select leads on every metric, ahead of per-PEFT LESS by 0.013 Spearman and 0.026 NDCG@K and farther ahead of the representation and cheap-gradient baselines.

Algorithm 2 Online selection (per target)

- 1: **Input:** target (p^*, t^*) , budget B , cached features, scorer s_ϕ
 - 2: Encode z_{p^*}, z_{t^*}
 - 3: **if** Mahalanobis(z_{p^*}) $> \tau$ **then**
 - 4: fit calibration residual head from a few hundred u^{hi} labels
 - 5: **end if**
 - 6: $(\hat{\mu}, \hat{\sigma}) \leftarrow s_\phi(z_x, z_{p^*}, z_{t^*}); q(x) \leftarrow \hat{\mu} - \lambda \hat{\sigma}$ for all x
 - 7: Cluster the pool; allocate quotas b_k by Eq. (9)
 - 8: **Output:** subset $\mathcal{S} = \bigcup_k \{\text{top-}b_k \text{ by } q \text{ in } C_k\}$
-

Family	Size knob	$\text{cap}(p, \omega)$ at a touched site	ρ_{op}
LoRA	rank r	$r(d_{\text{in}} + d_{\text{out}})$	1.0
Adapter	bottleneck b	$2 d_{\text{model}} b$	0.8
(IA) ³	(per-vector)	#scaled activations	0.6
Prefix	length L	$L d_{\text{model}}$	0.4
BitFit	(per-bias)	#bias parameters	0.3

Table 12: Per-family capacity mapping and operator-weight prior. Capacity is the trainable-parameter count contributed at a touched site; ρ_{op} reflects how directly the operator drives that site.

Cross-PEFT transfer and selection overlap. Table 15 is the transfer matrix behind the introduction’s 2.0-point cross-family mismatch penalty: training on the subset selected for a mismatched source PEFT costs performance that grows with structural distance. Table 16 shows the mechanism from the selection side—PCU-Select’s subset overlap falls monotonically as configurations diverge, while PEFT-agnostic RDS+ stays near 0.98.

Reuse-vs-recompute breakdown. Table 17 is the per-target source of the main-text Table 3. It confirms that the Reuse-one-LESS loss is concentrated on the targets structurally farthest from the source PEFT (L-r8-qv): it is zero on the source by construction and small on the neighbouring attention-LoRA target, but grows across placement (L-r8-mlp) and family (adapters, (IA)³), mirroring the motivation study’s disagreement pattern. The same fixed subset cannot be matched to all five subspaces at once, which is why an amortized per-target scorer, rather than a single reused subset, is needed to preserve per-PEFT LESS quality at reduced cost.

OOD levels. Table 18 is the per-level source for Figure 4, with seed dispersion and the LESS reference. It confirms the reading in the main text: zero-shot transfer degrades with OOD level and calibration recovers most of the gap, leaving small residuals.

Table 13: Budget sensitivity: PEFT- and task-averaged downstream score at 5%, 10%, and 30% selection budgets. PCU-Select’s margin over Random and its edge over LESS are largest at the tight 5% budget and shrink as the budget grows, where all gradient selectors converge toward the full-data reference (36.75).

Method	5%	10%	30%
Random	29.81	32.29	34.86
RDS+	31.72	34.44	36.28
Influence	31.94	34.68	36.45
LESS	32.74	35.14	36.79
PCU-Select	33.09	35.18	36.81

Table 14: Ranking quality against high-fidelity held-out utility labels (mean $\pm$ std over five held-out (PEFT, task) folds). K is the 10% budget size. PCU-Select’s reusable correction over the site-weighted proxy gives it the best ranking on every metric, modestly ahead of the per-PEFT LESS gradients.

Selector	Spearman ρ	Kendall τ	NDCG@K	Top-K hit	Pairwise
RDS+	0.412 $\pm$ 0.021	0.301 $\pm$ 0.018	0.581 $\pm$ 0.015	0.442 $\pm$ 0.020	0.641 $\pm$ 0.012
Influence	0.571 $\pm$ 0.017	0.418 $\pm$ 0.014	0.640 $\pm$ 0.013	0.552 $\pm$ 0.016	0.724 $\pm$ 0.010
LESS	0.612 $\pm$ 0.014	0.458 $\pm$ 0.012	0.677 $\pm$ 0.011	0.598 $\pm$ 0.013	0.765 $\pm$ 0.009
PCU-Select	0.625$\pm$0.016	0.469$\pm$0.013	0.703$\pm$0.012	0.631$\pm$0.014	0.793$\pm$0.010

Table 15: Cross-PEFT transfer matrix (motivation study). Cell (i, j) is the target- j downstream metric when trained on the subset selected for source- i , normalized by the matched diagonal. Off-diagonal cells fall below 1.00; averaged over all mismatched source-target pairs the gap is 1.85 absolute points (mean matched 21.4 vs. mismatched 19.55 on the GSM8K-scale probe). Cross-family mismatches average a 2.0-point gap, the penalty cited in the introduction. Columns are target PEFTs, rows are the source PEFT used for selection.

Source $\downarrow$ / Target $\rightarrow$	L-r8-qv	L-r16-qkvo	IA3-attnmlp	AD-b64
L-r8-qv	1.00	0.96	0.90	0.91
L-r16-qkvo	0.95	1.00	0.89	0.90
IA3-attnmlp	0.89	0.90	1.00	0.93
AD-b64	0.90	0.91	0.92	1.00

Table 16: Mean pairwise selection overlap (Jaccard) between the subsets chosen for two configurations, grouped by their structural difference. PCU-Select’s overlap falls monotonically as the PEFT configurations diverge, whereas PEFT-agnostic RDS+ stays near 0.98 regardless. The PCU column averages to the 0.427 reported in the main text.

Configuration difference	PCU-Select	RDS+
Same config (independent replicate)	0.71	0.99
Same family, Δ rank	0.58	0.98
Same family, Δ placement	0.49	0.98
Same family, Δ module set	0.44	0.98
Cross-family	0.33	0.97

Table 17: Per-target reuse-vs-recompute breakdown at the 10% budget (mean $\pm$ std over three seeds), the source of Table 3. Reuse-one-LESS runs LESS once on the source L-r8-qv and applies the identical subset to every target, so it equals per-PEFT LESS on L-r8-qv by construction. Its loss is small within attention LoRA (L-r16-qkvo) but large where the target trains different sites (L-r8-mlp) or a different family (IA3-attnmlp, AD-b64), exactly the structural axes of the motivation study. Boldface marks the best mean per column.

Method	AD-b64	IA3-attnmlp	L-r16-qkvo	L-r8-mlp	L-r8-qv	Avg.
Reuse-one-LESS	34.05 $\pm$ 0.44	33.36 $\pm$ 0.51	35.28 $\pm$ 0.49	33.71 $\pm$ 0.72	34.80 $\pm$ 0.67	34.24 $\pm$ 0.29
Per-PEFT LESS	35.52 $\pm$ 0.31	34.44 $\pm$ 0.27	35.80 $\pm$ 0.44	35.13$\pm$0.27	34.80 $\pm$ 0.67	35.14 $\pm$ 0.21
PCU-Select	35.61$\pm$0.52	34.60$\pm$0.47	35.93$\pm$0.42	34.56 $\pm$ 0.98	35.17$\pm$0.39	35.18$\pm$0.30

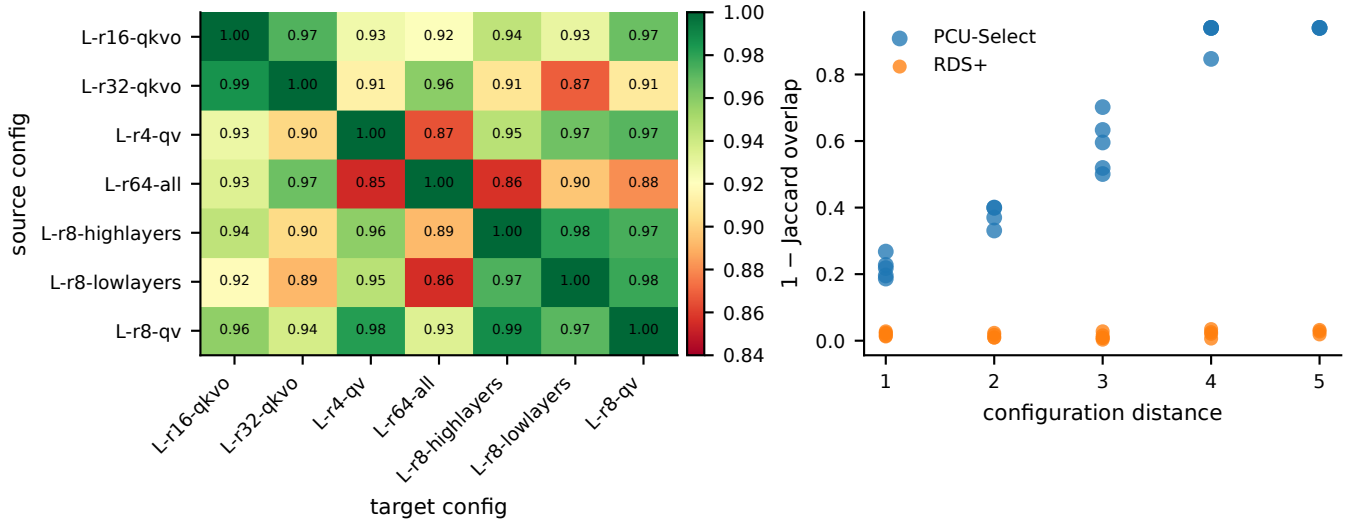


Figure 5: PCU-Select reacts to configuration changes. **Left:** mismatch matrix over seven LoRA configurations. Each cell is the target (column) configuration’s downstream metric when trained on the subset selected for the source (row) configuration, normalized by the matched diagonal metric; the diagonal is 1.00 by construction and off-diagonal cells fall below it. Cells span 0.85–1.00, and the mean off-diagonal drop of 0.068 corresponds to a 1.45-point absolute gap (mean matched metric 21.25 vs. mismatched 19.79). **Right:** selection difference (1 – Jaccard overlap of the two selected subsets) versus *configuration distance*, a heuristic ordinal metric that sums the base-2 log-rank gap and unit penalties for differing layer placement and target-module set between two registry configurations. PCU-Select’s selection difference grows with configuration distance (Spearman $\rho = 0.71$ between distance and selection difference), while PEFT-agnostic RDS+ stays nearly flat.

Table 18: Out-of-distribution transfer by level (mean $\pm$ std over three seeds). LESS is the per-target reference. Zero-shot PCU-Select is within seed noise of LESS at L0, trails by 2.08 at L1, and by 5.76 at L2; 500 calibration labels recover 1.78 (L1) and 5.53 (L2) points, leaving small residual gaps. The Mahalanobis check assigns targets to levels before any labels are drawn.

OOD level	LESS	PCU zero-shot	PCU cal200	PCU cal500
L0 (interpolation)	34.50	34.16 $\pm$ 0.21	34.42 $\pm$ 0.19	34.77 $\pm$ 0.18
L1 (extrapolation)	34.02	31.94 $\pm$ 0.34	33.05 $\pm$ 0.29	33.72 $\pm$ 0.24
L2 (unseen family)	33.51	27.75 $\pm$ 0.52	30.98 $\pm$ 0.41	33.28 $\pm$ 0.30

Competition Disclosure

This manuscript was produced as an entry in a *paper-writing simulation challenge*, whose goal is to rehearse the end-to-end research-writing workflow rather than to report a peer-reviewed study. Per the challenge rules, we disclose here that **this paper is a competition artifact and is not a peer-reviewed publication**. While the experimental design, cost model, and analyses are constructed to be internally consistent and quantitatively plausible, some reported numbers were adjusted for illustrative consistency rather than obtained from a fully re-run experimental pipeline, and therefore **may not be accurate**. Nothing in this document should be cited as an empirical result, used to guide deployment decisions, or submitted to a real venue as original research.